deterministic systems and would be expected to be only first order accurate. It could be made symmetric by a simple change, for example replacing $\boldsymbol{p}_n$ in the force computation at the first step by $\boldsymbol{p}_{n+1/2}$, but these changes would make the method implicit.

8.9.2.2 Shardlow's Splitting Method

Shardlow [333] proposed splitting methods for DPD based on dividing the force field of the DPD system into its separate (conservative+dissipative+random) parts. Several integrators are then derived based on approximating the various terms. In the most popular and efficient variant, the scheme of Brünger, Brooks and Karplus (BBK) [55] is used to solve the dissipative+stochastic terms, with standard velocity Verlet applied to the conservative part.

We first separate the system of stochastic differential equations for DPD (8.90) into three pieces, which we label as A, B and O:

$$\mathrm{d}\begin{bmatrix}\boldsymbol{q}_i\\ \boldsymbol{p}_i\end{bmatrix} = \underbrace{\begin{bmatrix} m_i^{-1}\boldsymbol{p}_i\\ \boldsymbol{0}\end{bmatrix}\mathrm{d}t}_{\mathrm{A}} + \underbrace{\begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{F}_i^{\mathrm{C}}\end{bmatrix}\mathrm{d}t}_{\mathrm{B}} + \underbrace{\begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{F}_i^{\mathrm{D}}\mathrm{d}t + \sigma\boldsymbol{R}_i\end{bmatrix}}_{\mathrm{O}}, \tag{8.94}$$

where $\boldsymbol{R}_i = \sum_{j\neq i}\omega^{\mathrm{R}}(r_{ij})\hat{\boldsymbol{q}}_{ij}\mathrm{dW}_{ij}$, for standard i.i.d. Wiener processes $\mathrm{W}_{ij}(t)$, such that $\mathrm{dW}_{ij} = \mathrm{dW}_{ji}$. The generators for each part of the SDE may be written out as follows:

$$\mathcal{L}_{\mathrm{A}} = \sum_i \frac{\boldsymbol{p}_i}{m_i}\cdot\nabla_{\boldsymbol{q}_i}, \tag{8.95}$$

$$\mathcal{L}_{\mathrm{B}} = \sum_i \boldsymbol{F}_i^{\mathrm{C}}\cdot\nabla_{\boldsymbol{p}_i} = -\sum_i \nabla_{\boldsymbol{q}_i}U(\boldsymbol{q})\cdot\nabla_{\boldsymbol{p}_i}, \tag{8.96}$$

$$\mathcal{L}_{\mathrm{O}} = \sum_i\sum_{j\neq i}\left(-\gamma\omega^{\mathrm{D}}(r_{ij})(\hat{\boldsymbol{q}}_{ij}\cdot\boldsymbol{v}_{ij}) + \frac{\sigma^2}{2}[\omega^{\mathrm{R}}(r_{ij})]^2\hat{\boldsymbol{q}}_{ij}\cdot\left[\nabla_{\boldsymbol{p}_i}-\nabla_{\boldsymbol{p}_j}\right]\right)\hat{\boldsymbol{q}}_{ij}\cdot\nabla_{\boldsymbol{p}_i}. \tag{8.97}$$

Thus, the generator for the DPD system can be written as

$$\mathcal{L}_{\mathrm{DPD}} = \mathcal{L}_{\mathrm{A}} + \mathcal{L}_{\mathrm{B}} + \mathcal{L}_{\mathrm{O}}, \tag{8.98}$$

and the flow map may be written in the standard shorthand as

$$\mathscr{F}_t = e^{t\mathcal{L}_{\mathrm{DPD}}}.$$

Approximations of $\mathscr{F}_t$ can be constructed as for Langevin dynamics.